

BILINEAR AND QUADRATIC FORMS

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1. TRACE MAPS

Definition 1.0.1. (♠ TODO: AI generated, verify) Let R be a (not necessarily commutative) ring, and let $n \in \mathbb{N}_{\geq 1}$. For any square matrix $A = (a_{ij}) \in M_{n,n}(R)$, the *trace of A* is the element

$$\mathrm{tr}(A) = \sum_{i=1}^n a_{ii}$$

in R .

The trace map $\mathrm{tr} : M_{n,n}(R) \rightarrow R$ is additive: $\mathrm{tr}(A + B) = \mathrm{tr}(A) + \mathrm{tr}(B)$ for all $A, B \in M_{n,n}(R)$. If R is commutative, then the trace satisfies the cyclic property

$$\mathrm{tr}(AB) = \mathrm{tr}(BA)$$

for all $A, B \in M_{n,n}(R)$, and more generally $\mathrm{tr}(A_1 A_2 \cdots A_k) = \mathrm{tr}(A_2 \cdots A_k A_1)$ for any cyclic permutation of a product.

Definition 1.0.2. Let R be a ring. Let A be an R -algebra. A *trace map on A* is any R -module homomorphism

$$\tau : A \rightarrow T$$

that has the “basic trace property”, i.e. that

$$\tau(\mu\alpha) = \tau(\alpha\mu)$$

for all $\mu, \alpha \in A$.

In particular, note that the notion of trace map on $\mathrm{End}_R(M)$ makes sense.

Definition 1.0.3. Let R be a ring. Let A be an R -algebra.

1. The *(universal) trace map on A* is the canonical projection map

$$\mathrm{Tr}_A : A \rightarrow A/[A, A].$$

It is universal amongst the trace maps on A in the sense of Definition 1.0.2, i.e. any trace map

$$\tau : A \rightarrow T,$$

uniquely factors through an R -module homomorphism $A/[A, A] \rightarrow T$:

$$\begin{array}{ccc} A & \xrightarrow{\text{Tr}_M} & A/[A, A] \\ & \searrow \tau & \downarrow \exists! \bar{\tau} \\ & & T \end{array}$$

2. Given an R -module M , the universal trace map on $\text{End}_R(M)$ may be denoted by Tr_M .

Definition 1.0.4. Let R be a ring. Let M be an R -module. Let $M^\vee = \text{Hom}_R(M, R)$ be the dual of M .

1. In the case that M is a left R -module, the *evaluation map* is the abelian group homomorphism

$$\text{ev} : M^\vee \otimes_R M \rightarrow R$$

given on pure tensors by

$$\text{ev}(f \otimes m) = f(m).$$

2. In the case that M is a right R -module, the *evaluation map* is the abelian group homomorphism

$$\text{ev} : M \otimes_R M^\vee \rightarrow R$$

given on pure tensors by

$$\text{ev}(m \otimes f) = f(m).$$

Definition 1.0.5. Let R be a ring. Let M be an (left) R -module. The *trace ideal of M* , denoted by notations such as $\tau_R(M)$ or $\text{Tr}_R(M)$, is the R -ideal of R generated by the image of the evaluation homomorphism (Definition 1.0.4)

$$\text{ev} : M^\vee \otimes_R M \rightarrow R.$$

Definition 1.0.6. Let R be a ring. Let M be an R -module. Let $M^\vee = \text{Hom}_R(M, R)$ be the dual of M .

1. In the case that M is a left R -module, there is a *canonical evaluation map* or a *canonical map*

$$\Phi : M^\vee \otimes_R M \rightarrow \text{End}_R(M)$$

which maps a pure tensor $f \otimes m$ to the endomorphism $x \mapsto f(x)m$ of M . In fact, these canonical evaluation maps form a natural transformation

$$(M \mapsto M^\vee \otimes_R M) \Rightarrow (M \mapsto \text{End}_R(M)) : {}_R \text{Mod} \rightarrow \text{Ab},$$

which may be called the *evaluation natural transformation*.

If M is an (R, S) -bimodule for a ring S , then the canonical map in fact has a right S -module structure.

2. In the case that M is a right R -module, there is a **canonical evaluation map** or a **canonical map**

$$\Phi : M \otimes_R M^\vee \rightarrow \text{End}_R(M)$$

which maps a pure tensor $m \otimes f$ to the endomorphism $x \mapsto mf(x)$ of M . In fact, these canonical evaluation maps form a natural transformation

$$(M \mapsto M \otimes_R M^\vee) \Rightarrow (M \mapsto \text{End}_R(M)) : \text{Mod}_R \rightarrow \text{Ab},$$

which may be called the **evaluation natural transformation**.

If M is an (S, R) -bimodule for a ring S , then the canonical map in fact has a left S -module structure.

Proposition 1.0.7. Let R be a ring. Let P be a projective, finitely-generated (left) R -module. The canonical evaluation map (Definition 1.0.6)

$$\Phi : P^\vee \otimes_R P \rightarrow \text{End}_R(P)$$

is an isomorphism.

Proof. By ??, a module over a ring is projective if and only if it is a direct summand of a free module. In the case that P is a finitely generated free module $R^{\oplus n}$, (♠ TODO: continue) $\square$

Definition 1.0.8. Let R be a ring. Let P be a projective, finitely-generated (left) R -module.

Recall that the canonical evaluation morphism (Definition 1.0.6)

$$\Phi : P^\vee \otimes_R P \rightarrow \text{End}_R(P)$$

is an isomorphism (Proposition 1.0.7). The **canonical trace map for P** is the composition

$$\text{Tr}_P : \text{End}_R(P) \xrightarrow{\Phi^{-1}} P^\vee \otimes_R P \xrightarrow{\text{ev}} R$$

where ev is the evaluation map (Definition 1.0.4).

Lemma 1.0.9. Let R be a ring. Let P be a projective, finitely-generated (left) R -module.

The canonical trace map (Definition 1.0.8)

$$\text{Tr}_P : \text{End}_R(P) \rightarrow R$$

for P is a trace map (Definition 1.0.2) on $\text{End}_R(P)$. (♠ TODO: prove)

Proposition 1.0.10. Let R be a ring. Let P be a projective, finitely-generated (left) R -module.

Given $\phi \in \text{End}_R(P)$, its image under the canonical trace map (Definition 1.0.8)

$$\text{Tr}_P : \text{End}_R(P) \rightarrow R$$

for P can be computed as follows:

1. Choose a module Q , an integer $n \geq 1$, and an isomorphism $R^{\oplus n} \cong P \oplus Q$.
2. Extend $\phi : P \rightarrow P$ to $\tilde{\phi} : R^n \rightarrow R^n$ by letting $\tilde{\phi}$ be 0 on Q .
3. $\text{Tr}_P(\phi)$ equals the trace of the endomorphism $\tilde{\phi}$, which may in turn be regarded as the trace of a matrix (Definition 1.0.1) representing $\tilde{\phi}$. (♠ TODO: this trace)

(♠ TODO: prove)